

Analysis of Transistor-level CMOS Inverter in LTspice

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1 Design Parameters & Tool Setup

1.1 Parameters

The NMOS and PMOS transistors are modeled using the typical values for 180nm CMOS Small-Signal Model [1]. The SPICE directives for the NMOS and PMOS transistors are as follows:

- **NMOS:** `.model NMOS_180nm NMOS (LEVEL=1 VT0=0.8 KP=90u GAMMA=0.5 PHI=0.68 LAMBDA=0.06 TOX=0.02u CJ=2.4e-4 CJSW=2.0e-10 CGSO=2.0e-10 CGDO=2.0e-10 PB=0.9 NSUB=6e15)`
- **PMOS:** `.model PMOS_180nm PMOS (LEVEL=1 VT0=-0.9 KP=30u GAMMA=0.5 PHI=0.68 LAMBDA=0.06 TOX=0.02u CJ=2.4e-4 CJSW=2.0e-10 CGSO=2.0e-10 CGDO=2.0e-10 PB=0.9 NSUB=6e15)`

1.2 LTspice Schematic

```
.model NMOS_ NMOS(Vto=0.1 Kp=1.73m Lambda=0.027 Rd=2 Rs=2  
+ Cbd=1e-14 Cbs=1e-14 Cgso=1.5e-7 Cgdo=6.2e-9 Cgbo=1e-9 Tox=2e-8)  
.model PMOS_ PMOS(Vto=-0.8 Kp=1.78m Lambda=0.022 Rd=1 Rs=1  
+ Cbd=1e-14 Cbs=1e-14 Cgso=6.066e-9 Cgdo=6.066e-9 Cgbo=1 Tox=9.7e-9)
```

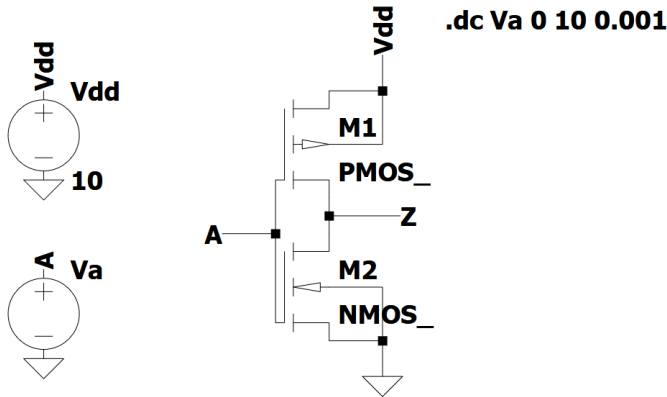


Figure 1: Circuit setup of the transistor-level CMOS inverter in LTspice.

From Fig.1, the circuit setup of the transistor-level CMOS inverter in LTspice is shown. First, on the left side, there are two voltage sources, the power supply voltage (Vdd) is set to 10V , while V2 represents the input voltage (A). Then, on the right side, there are two MOSFETs used to represent the PMOS and NMOS transistors. The upper side is the PMOS transistor, which

is connected to Vdd, while the lower side is the NMOS transistor, which is connected to the ground. Finally, the gate of both transistors is connected to A and the output voltage (Z) is connected to the junction between the two transistors. [1]

2 DC Analysis (Voltage Transfer Characteristics)

2.1 VTC Curve Generation

To obtain the Voltage Transfer Characteristic (VTC), a DC sweep simulation is performed, sweeping the input voltage from 0V to 1.8V with a step size of 0.001.

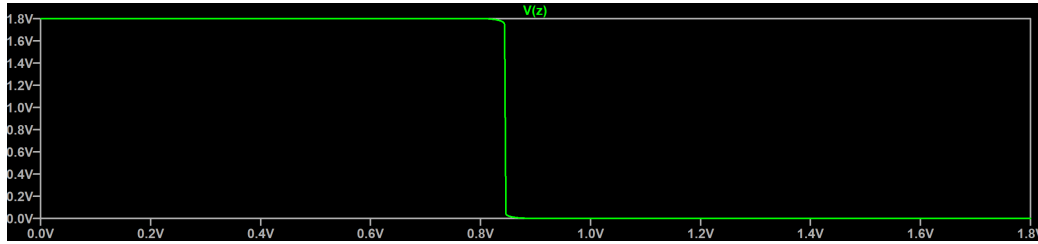


Figure 2: Voltage Transfer Characteristic (VTC) of the CMOS Inverter.

From Fig.2, the relationship between the V_a and V_z is shown. It was found that when V_a was lower than 0.8V, the output voltage (V_z) was high, and when V_a was higher than 0.9V, V_z was low.

2.2 Critical Voltages

From the VTC curve, the critical voltages are extracted (where the slope is -1):

- $V_{OH} = 1.8 \text{ V}$
- $V_{OL} = 0 \text{ V}$
- $V_{IH} = 0.857 \text{ V}$
- $V_{IL} = 0.829 \text{ V}$

2.3 Noise Margins

The noise margins are calculated using the extracted critical voltages:

- High Noise Margin ($NM_H = V_{OH} - V_{IH}$) = 0.943 V
- Low Noise Margin ($NM_L = V_{IL} - V_{OL}$) = 0.829 V

2.4 Switching Threshold

The switching threshold (V_M), where $V_{in} = V_{out}$, is measured from the VTC:

- $V_M = 0.845 \text{ V}$

3 Transient Analysis (Timing Characteristics)

3.1 Transient Simulation Setup

To analyze the timing characteristics, a transient simulation is performed. The input is driven by a pulse voltage source with the following parameters:

- $V_{initial} = 0 \text{ V}$, $V_{on} = 1.8 \text{ V}$
- $T_{delay} = 0 \text{ s}$, $T_{rise} = 1 \text{ ns}$, $T_{fall} = 1 \text{ ns}$
- $T_{on} = 10 \text{ ns}$, $T_{period} = 20 \text{ ns}$

3.2 Propagation Delay and Timing

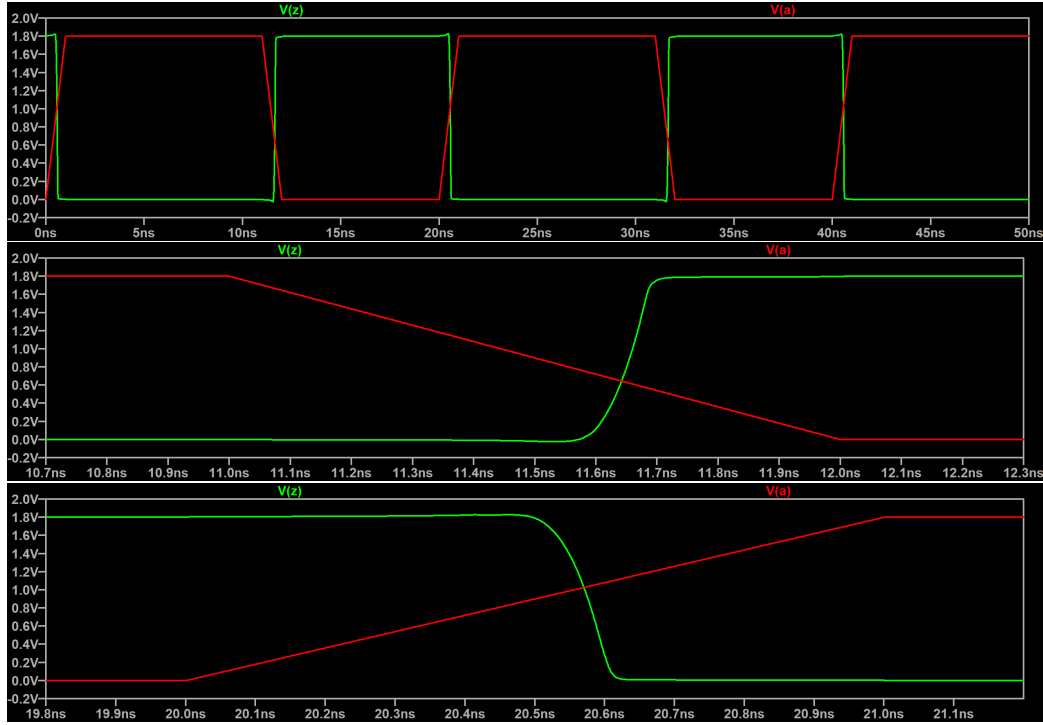


Figure 3: Top: Transient response of the CMOS inverter. Middle: Zoomed-in view of the rise time. Bottom: Zoomed-in view of the fall time.

The propagation delay is measured between the 50% points of the input and output transitions:

- High-to-Low Delay (t_{pHL}) = 7.5 ps

- Low-to-High Delay (t_{pLH}) = 150 ps
- Average Delay (t_p) = 79 ps

The rise and fall times are measured between the 10% and 90% points of the output signal:

- Rise Time (t_r) = 74 ps
- Fall Time (t_f) = 72 ps

References

- [1] Scribd. 180nm process typical parameter values.
<https://www.scribd.com/presentation/470685433/180nm-process-typical-parameter-values>.